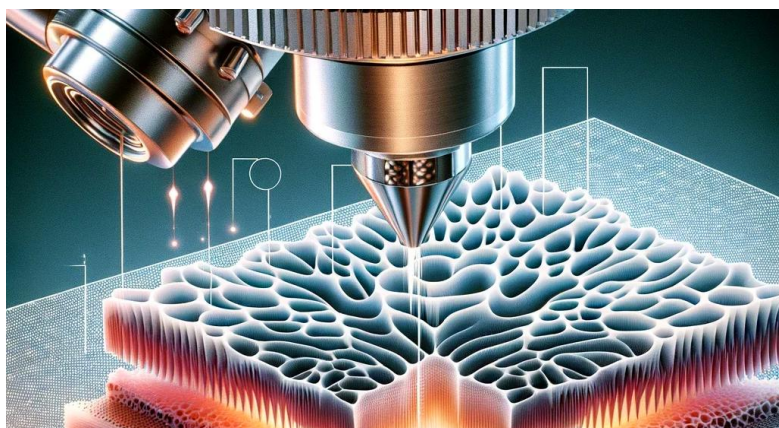




Additive artistry

增材技术

难度评级：★★★★☆☆

Electronics can now be printed onto living **tissues**

电子元件也可以打印至活体组织

【 Para.1 】

It began in the 1980s as a way to quickly produce simple **prototype** models in plastic. These days 3D printing—or “additive manufacturing”, to give it its **posher** name—is used to make kitchen **gadgets**, jet-engine parts, dental **implants** and even some buildings.

3D打印始于20世纪80年代，最初用于快速制作简单的塑料原型模型；如今，这项技术——或者用更时髦的说法，即“增材制造”——已被用于制造厨房小器具、喷气发动机零部件、牙科植入物，甚至某些建筑物。

【 Para.2 】

In a paper published in Science Advances, a team of researchers led by Yong Lin Kong and his colleagues at

文章导读：

3D打印技术应用领域宽广。有研究人员开发出新的应用前景，可以将电子电路打印在活体组织上。3D打印电路并不稀奇。但打印至活体组织需要用到特殊的电子墨水，并为此改良了退火工艺。同时，他们还制造了一款微型探头，可以调节墨水的微观结构。这项技术可以用于电子实时监测移植手术后患者的恢复状况，同时还可以帮助精准用药。

additive n. /'ædətɪv/ a substance that is added in small amounts to sth, especially food, in order to improve it, give it colour, make it last longer, etc. (尤指食品的) 添加剂, 添加物

food **additives** 食品添加剂

artistry n. /'ɑ:tɪstri/ [U] the skill of an artist 艺术技巧

He played the piece with effortless **artistry**.

他游刃有余地演奏了这首乐曲。

electronic a. /ɪˌlek'trɒnɪk/ [usually before noun] (of a device 装置) having or using many small parts, such as microchips, that control and direct a small electric current 电子的；电子器件的【考研大纲词汇】

an **electronic calculator** 电子计算器

tissue n. [U] a collection of cells that form the different parts of humans, animals and plants (人、动植物细胞的) 组织【考研大纲词汇】

muscle/brain/nerve, etc. **tissue** 肌肉、大脑、神经等组织

prototype n. /'prəʊtətaɪp/ ~ (for/of sth) the first design of sth from which other forms are copied or developed 原型；雏形；最初形态【考研大纲词汇】

the **prototype** of the modern bicycle 现代自行车的雏形

posh a. (informal) elegant and expensive 优雅豪华的；富丽堂皇的

a **posh hotel** 豪华旅馆

gadget n. /'gædʒɪt/ a small tool or device that does sth useful 小器具；小装置

implant n. /'ɪmplɑ:nt/ something that is put into a person's body in a medical operation (植入人体中的) 移植

silicone breast **implants** 硅酮乳房填充物

come up with phrase [no passive] to find or produce an answer, a sum of money, etc. 找到（答案）；拿出（一笔钱等）【考研大纲词汇】

She **came up with** a new idea for increasing sales.

她想出了增加销售量的一个新招儿。

Rice University in Houston, Texas, propose another application. They have **come up with** a way to print electronic **circuits** onto delicate materials, including living tissue. The researchers hope that could eventually lead to medical implants that wirelessly monitor the condition of their recipients, high-tech pills and more.

得克萨斯州休斯顿莱斯大学的Yong Lin Kong及其同事领导的研究团队在《科学进展》（Science Advances）上发表了一篇论文，提出了一种新的应用。他们开发出一种方法，可以将电子电路打印在包括活体组织在内的脆弱材料上。研究人员希望，这最终能带来可无线监测受体状况的医疗植入物、高科技药丸等产品。

【 Para.3 】

3D-printing circuitry is not a new idea. Nano Dimension, based in Massachusetts, for instance, prints electronics into communications equipment and medical sensors. Printers made by Optomec, based in New Mexico, build **antennas** directly into smartphones cases, as well as printing circuits onto touch screens and solar panels.

3D打印电路并非新概念。例如，总部位于马萨诸塞州的Nano Dimension公司，就将电子元件打印到通信设备和医疗传感器中。总部位于新墨西哥州的Optomec公司生产的打印机，不仅能将天线直接集成到智能手机外壳中，还能在触摸屏和太阳能电池板上打印电路。

【 Para.4 】

The process uses electronic inks that contain tiny particles of conductive material, such as **copper**, silver or gold. The

circuit n. /'s3:kt/ the complete path of wires and equipment along which an electric current flows 电路；线路【考研大纲词汇】

an electrical **circuit** 电路

antenna n. = aerial An aerial is a device or a piece of wire that sends and receives television or radio signals and is usually attached to a radio, television, car, or building. 天线【考研大纲词汇】

radio **antennas** 收音机天线

copper n. Copper is reddish brown metal that is used to make things such as coins and electrical wires. 铜【考研大纲词汇】

Chile is the world's largest producer of **copper**.

智利是世界上最大的产铜国。

fuse v. When things fuse or are fused, they join together physically or chemically, usually to become one thing. You can also say that one thing fuses with another. 结合；融合；熔合【考研大纲词汇】

The skull bones **fuse** between the ages of fifteen and twenty-five. 颅骨在15和25岁之间长合。

anneal v. /ə'ni:l/ to temper or toughen (something) by heat treatment 使退火

furnace n. A furnace is a container or enclosed space in which a very hot fire is made, for example, to melt metal, burn rubbish, or produce heat for a building or house. 熔炉；火炉【考研大纲词汇】

zap v. (informal) ~ sb/sth (with sth) to destroy, kill or hit sb/sth suddenly and with force (突然而猛烈地) 毁坏，杀死，打击

The monster got **zapped** by a flying saucer (= in a computer game). 怪兽被飞碟杀死了。

polymer n. /'pɒlɪmə(r)/ (chemistry 化) a natural or artificial substance consisting of large molecules (= groups of atoms) that are made from combinations of small simple molecules 聚合物；多聚体

beam n. /bi:m/ a line of light, electric waves or particles 光线；（电波的）波束；（粒子的）束【考研大纲词汇】

narrow **beams** of light/sunlight 一丝丝的光线/阳光

precision n. /pri'sɪzn/ [U] the quality of being exact, accurate and careful 精确；准确；细致【考研大纲词汇】

SYN accuracy

done with mathematical **precision** 以数学般的精确完成的

substance n. /'sʌbstəns/ [C] a type of solid, liquid or gas that has particular qualities 物质；物品；东西【考研大纲词汇】

a chemical/radioactive, etc. **substance** 化学、放射性等物质

resonator n. /'rezənətə(r)/ (technical 术语) a device for making sound louder and stronger, especially in a musical instrument 共振器；共鸣箱

particles are then **fused** together to form a conductive pathway, usually by a process known as **annealing**, which involves either heating the entire product in a **furnace**, or **zapping** the inks with a laser.

这项技术使用的是含有微小导电材料颗粒（如铜、银或金）的电子墨水。随后，这些颗粒会通过一种被称为“退火”的工艺被熔合在一起，形成导电路径。所谓退火，一般涉及将整个产品放入炉中加热，或者用激光对准墨水照射。

【 Para.5 】

But that approach has limits, says Dr Kong. The heat needed to fuse the particles can damage sensitive materials, including some **polymers** and living tissue. It is this issue that Dr Kong and his colleagues now think they have overcome. They have developed a way to anneal electronic inks using microwaves focused into a **beam** smaller than a human hair. The **precision** of the beam ensures that only the ink particles heat up, avoiding any damage to the surrounding **substances**.

但孔博士指出，这种方法存在局限性。熔合颗粒所需的热量可能会损坏敏感材料，包括某些聚合物和活体组织。孔博士及其同事认为，这一问题现已被他们攻克。他们开发了一种利用微波对电子墨水进行退火的方法，将微波聚焦成比人类头发还细的微束。该微束的高精度确保只有墨水颗粒受热，而不会损害周围的物质。

【 Para.6 】

The researchers produced a special **resonator** to **amplify**

amplify v. /'æmplɪfaɪ/ to increase sth in strength, especially sound 放大，增强（声音等）【考研大纲词汇】
to **amplify** a guitar/an electric current/a signal
放大吉他声音/电流/信号

taper v. /'teɪpə(r)/ to become gradually narrower; to make sth become gradually narrower (使) 逐渐变窄
The tail **tapered** to a rounded tip.
尾部越来越细，最后成了个圆尖。

compact a. smaller than is usual for things of the same kind 小型的；袖珍的【考研大纲词汇】
a **compact** camera 袖珍照相机

valve n. a structure in the heart or in a vein that lets blood flow in one direction only (心脏或血管的) 瓣(膜)【考研大纲词汇】

trachea n. /trə'ki:ə/ (anatomy 解) the tube in the throat that carries air to the lungs 气管
SYN windpipe

graft n. /grɑ:ft/ a piece of skin, bone, etc. removed from a living body and placed in another part of the body which has been damaged; the process or result of doing this 移植的皮肤(或骨骼等)；移植
a **skin graft** 皮移植片

strain n. the pressure that is put on sth when a physical force stretches, pushes, or pulls it 压力；拉力；张力；应力【考研大纲词汇】
The rope broke under the **strain**. 绳子给拉断了。

gauge n. /geɪdʒ/ (often in compounds 常构成复合词) an instrument for measuring the amount or level of sth 测量仪器(或仪表)；计量器【考研大纲词汇】
a fuel/petrol/temperature, etc. **gauge** 燃料表、汽油量表、温度计等

femur n. /'fi:mə(r)/ (anatomy 解) the thigh bone 股骨

ingestible a. /ɪn'dʒestəbl/ 可吸收的；可摄取的
ingest v. /ɪn'dʒest/ (technical 术语) to take food, drugs, etc. into your body, usually by swallowing 摄入；食入；咽下

computerised a. a computerized system, process, or business is one in which the work is done by computer. 计算机化的【考研大纲词汇】
The National Cancer Institute now has a **computerized** system that can quickly provide information.
国家癌症研究所现在拥有一套能快速提供信息的计算机化系统。

relay v. to receive and send on information, news, etc. to sb 接转，转送，转发(信息、消息等)【考研大纲词汇】
He **relayed** the message to his boss.
他将这个消息转给了他的老板。

canvas n. /'kænvəs/ A canvas is a piece of canvas or similar material on which an oil painting can be done. 油画布【考研大纲词汇】

the microwave energy and a **tapered** tip to focus the beam into a dot less than 200 millionths of a metre across. The kit is **compact** enough to be fitted to a desktop 3D printer. By adjusting the power of the beam, it is possible to change the microstructure of the ink. This can be used to create regions of higher or lower conductivity along the circuit, allowing some components, such as resistors, to be created within the wiring.

研究人员还制造了一个用于放大微波能量的特殊谐振器，以及一个将微波束聚焦成直径小于2亿分之一米微点的锥形探头。该装置足够袖珍，可安装在台式3D打印机上。调节微波束的功率，可以改变墨水的微观结构。这可用于沿电路形成导电性更高或更低的区域，从而在布线中制造出电阻器等元件。

【 Para.7 】

Dr Kong and his colleagues have plenty of applications in mind. 3D printing has already been used to make biological tissues for transplant, including heart **valves** and **tracheas**, without the need for human or animal donors. Adding circuitry could turn such **grafts** into wireless sensors, allowing them to monitor a patient's progress in real time, or even stimulate the activity of the organ they are printed on.

孔博士及其同事构想了许多应用场景。3D打印技术已被用于制造用于移植的生物组织，包括心脏瓣膜和气管，无需人类或动物供体。若加入电路，此类移植组织便可转变为无线传感器，从而实时监测患者的恢复情况，甚至能刺激其所附着器官的活动。

➤ 今日句子分析

主句

The particles are then fused together

主语

谓语

to form a conductive pathway, usually by a process

目的状语

方式状语

known as annealing, which involves

定语

主语 谓语

either heating the entire product in a furnace, or zapping the inks with a laser.

宾语

句子分析：

主句：

主语-The particles；谓语-are then fused together；目的状语-to form a conductive pathway；方式状语-usually by a process；后置定语-known as annealing；

定语从句（这一部分其实都是在解释“什么是退火”，因此在翻译时可以考虑另起一句，对其进行解释）：

从句主语-which；谓语-involves（“包含；包括”，这个动词有时也可以理解为“需要……”，例：The job involves me travelling all over the country. 这份工作需要我在全各地跑。）；宾语1-either heating the entire product in a furnace；（连词）-or；宾语2-zapping the inks with a laser（zap一般指“摧毁”，这里结合后面的laser，可以理解为“用激光集中照射……”，与前面的heating the entire...进行区别）。

【 Para.8 】

As a proof of concept, the researchers have printed a **strain gauge** onto both a cow's **femur** and the sorts of polymers used in replacement knees and hips for human patients. Another idea is printing circuits onto "**ingestible** diagnostic devices". Essentially **computerised** pills, these are designed to **relay** information from inside a patient or carefully control the release of drugs. The researchers have even managed to print a humidity sensor onto a living leaf. The natural world, it seems, is the next **canvas** for additive artists.

为了印证这一概念，研究人员已在牛股骨以及用于人类膝关节和髋关节置换的聚合物材料上打印了应变片。另一个构想是将电路打印在“可吞服诊断设备”上。这些设备本质上是“电子化药丸”，旨在从患者体内传输信息或精确控制药物的释放。研究人员甚至成功地将一个湿度传感器打印在活叶片上。看来，自然界将成为增材制造艺术家的下一块创作画布。

➤ 写作积累

Para.7 Dr Kong and his colleagues have plenty of applications in mind. have sth in mind 字面意思“心中已有某个想法、计划、目标、人选或事物”或者“正在考虑或打算做某事”。文中的 have plenty of applications in mind 因而并不是指“脑子里记着很多应用”，而是指“他们已经构思好了很多应用方向，但尚未付诸实现或公开发布”。

与 mind 相关，但用法略有不同的表达还包括：

(have sth) on one's mind “担忧/记挂……”，例：You've been on my mind all day. 我一整天都在为你担心。

to one's mind “依……之见”，例：It was a ridiculous thing to do, to my mind. 依我看，这样做是很荒唐的。

Para.8 Essentially computerised pills, these are designed to relay information from inside a patient or carefully control the release of drugs.

Essentially computerised pills 可视为主语 these 的同位语（但是是前置，有一说则认为这种结构相当于 being sth，本质上是作状语），对它进行定义、解释或归类。文中这个短语是用来强调“这些东西本质上是电子化的药丸”。

写作中时需要注意前置的同位语必须与主句主语指代同一事物，否则会造成逻辑混乱。

仿写：

该模式本质上是订阅制服务，旨在产生经常性收入同时保持客户忠诚度。

Essentially a subscription-based service, this model is designed to generate recurring revenue while retaining customer loyalty.

区分：

This model is essentially a subscription-based service, designed to generate recurring revenue while retaining customer loyalty.